

Data Visualization Practical Documentation

Lesson 9: Cleaning and Structuring Data for Analysis | Tableau Desktop Practical Steps

Step 1: Data Cleaning & Pivoting (Sections A & B)

CALCULATED FIELDS:

- **Physics_Clean:** IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)
- **Attendance_Clean:** FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))
- **Math_Clean:** IFNULL([Math], 0)
- **Chemistry_Clean:** IFNULL([Chemistry], 0)

ACTIONS:

- Navigate to Data Source tab -> Select columns Math, Physics, Chemistry, Programming -> Right-click -> Pivot.
- Rename 'Pivot Field Names' to Subject and 'Pivot Field Values' to Score (set data type to Number decimal).

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source loaded. The 'Connections' pane on the left shows 'Sem1_Scores' as a text file. The 'Fields' pane on the left lists various calculated fields. The main view displays a table with columns for Student ID, Full Name, Dept Info, and scores for Math, Physics, Chemistry, and Programming. The scores are displayed as percentages.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %
S101	Aarav Sharma	CSE-01	88	0.920000	90	98	88 %
S102	Priya Verma	ECE-02	79	0.780000	82	88	92 %
S103	Rohan Gupta	CSE-01	82	0.850000	80	90	80 %
S104	Sneha Patel	IT-03	95	0.900000	94	98	96 %
S105	Aditya Rao	MECH-04	52	0.580000	55	65	70 %
S106	Ananya Singh	CSE-01	90	0.940000	88	95	90 %
S107	Rahul Nair	ECE-02	74	0.730000	78	82	75 %
S108	Kavya Reddy	IT-03	96	0.960000	95	99	99 %

Step 2: Multi-File Union & Cross-Database Joins (Sections C & D)

Semester:

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```

ACTIONS:

- Drag New Union onto canvas -> add Sem1_Scores.csv and Sem2_Scores.csv (combines to 20 total rows).
- Drag and join Department_Master.csv on Dept_Info = Dept_Code.
- Drag and join Student_Activities.csv on Student_ID = Student_ID.

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Sem1_Scores' as a text file. The 'Files' pane shows a list of files including 'Department_Master.csv', 'Sem1_Scores.csv', 'Sem2_Scores.csv', and 'Student_Activities.csv'. The main canvas displays a 'Union' of 'Sem1_Scores' and 'Sem2_Scores'. Below the canvas, the 'Field Properties' pane shows the 'Union' field with a dropdown menu set to 'Department_Master.csv'. The 'Performance Options' pane is also visible. The data table at the bottom shows 20 rows of data with columns: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, and Attendant.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendant
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %

Step 3: Scorecard & Formatting (Section E)

CALCULATED FIELDS:

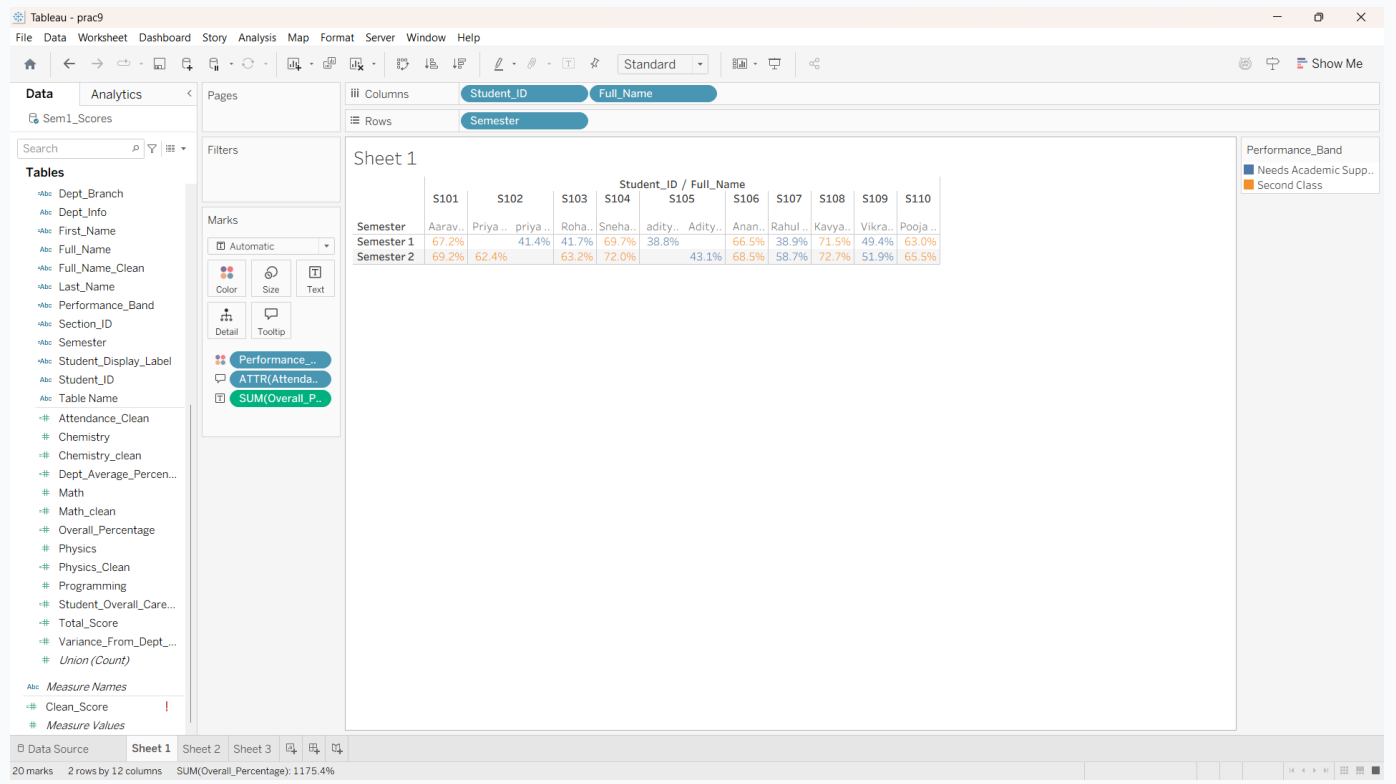
- **Total_Score:** [Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]
- **Overall_Percentage:** [Total_Score] / 400.0 (Set Default Number Format -> Percentage, 1 decimal)
- **Attendance_Status:** IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END

Performance_Band:

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END
```

ACTIONS:

- In Sheet 1, place Student_ID and Full_Name on Rows shelf, Semester on Columns shelf.
- Drag Overall_Percentage onto Text and Performance_Band onto Color.



Step 4: Level of Detail (LOD) Department Benchmarks (Section F)

CALCULATED FIELDS:

- **Dept_Average_Percentage_LOD:** { FIXED [Dept_Info] : AVG([Overall_Percentage]) }
- **Student_Overall_Career_Average:** { FIXED [Student_ID] : AVG([Overall_Percentage]) }
- **Variance_From_Dept_Average:** [Overall_Percentage] - [Dept_Average_Percentage_LOD]

Benchmark_Status:

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"

ELSE "Below Department Benchmark"

END

ACTIONS:

- In Sheet 2, place Dept_Info, Student_ID, and Full_Name on Rows shelf, Semester on Columns shelf.
- Place Overall_Percentage on Text, Benchmark_Status on Color, and Dept_Average_Percentage_LOD on Tooltip.

The screenshot shows the Tableau Desktop interface with a calculated field table in Sheet 2. The table displays student performance data categorized by department and semester, with color-coded benchmark status and tooltip values.

Dept_Info	Student..	Full_Name	Seme..	Seme..
CSE-01	S101	Aarav Sharma	67.2%	69.2%
	S103	Rohan Gupta	41.7%	63.2%
	S106	Ananya Singh	66.5%	68.5%
	S109	Vikram Joshi	49.4%	51.9%
ECE-02	S102	Priya Verma		62.4%
		priya VERMA	41.4%	
	S107	Rahul Nair	38.9%	58.7%
IT-03	S110	Pooja Iyer	63.0%	65.5%
	S104	Sneha Patel	69.7%	72.0%
	S108	Kavya Reddy	71.5%	72.7%
MECH-04	S105	Aditya Rao		43.1%
		aditya Rao	38.8%	

The interface also shows the 'Marks' card with 'Color' set to 'Benchmark_Status' and 'Detail' set to 'SUM(Dept_Ave...)' and 'SUM(Overall_P...)'.

Step 5: String Parsing & Final View (Section G)

CALCULATED FIELDS:

- **Full_Name_Clean:** `TRIM([Full_Name])`
- **Dept_Branch:** `SPLIT([Dept_Info], "-", 1)`
- **Section_ID:** `SPLIT([Dept_Info], "-", 2)`
- **Student_Display_Label:** `[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"`

ACTIONS:

- In Sheet 3, place Student_Display_Label on Rows shelf, Semester on Columns shelf.
- Place Dept_Branch and Section_ID on Filters shelf.
- Place Overall_Percentage on Text and Performance_Band on Color.

